Justin Duong

Atlanta, Georgia | (404)563-3347|Justinduong13@gmail.com| US citizen | LinkedIn

Objective

Computer engineering major with strong problem-solving skills and disruptive mindset specializing in controls system and additive manufacturing. Diligent and comfortable in working in high-stress environments, troubleshooting complex mechanical and software systems and collaborating with hardware and software teams to develop better solutions. Seeking a technical internship for Summer 2026.

Education

Georgia Institute of Technology | Atlanta, GA *August 2025 – Present*
Bachelor of Science in Computer Engineering, GPA 4.00 Expected Graduation, May 2028

Skills

Programming: Java, Python, C/C++, JavaScript, MATLAB, ROS2
Platforms: Linux (Ubuntu, Debian),
Hardware: Raspberry Pi, Arduino, ESP32, oscilloscope, logic analyzer, FDM/SLA 3D printing, laser cutting, CNC, basic machining
Software: Android Studio, SolidWorks, AutoCAD, Onshape, OpenCV, GitHub
Professional Organizations: Georgia Tech Institute of Electrical and Electronics Engineer
Communication: Design proposals, technical reports, presentations (large and small audiences)
Languages: English (fluent), Cantonese (fluent), Mandarin (fluent)

Experience

Convexity Electronics | Atlanta, GA *September 2025 – Present*
Startup Engineering Intern

- Analyzed composite spectral decompositions and experimentally evaluated new material mixtures to improve thermal conductivity and overall material performance
- Designed, prototyped, and mechanically debugged 5 iterations of a FDM 3D printer toolhead swapper system using CAD and rapid fabrication
- Integrated mechanical assemblies with motors, sensors, and control firmware using gcode
- Collected, managed, and interpreted engineering data related to material properties, supporting iterative design and validation processes

Georgia Tech Crablab Research | Atlanta, GA *September 2025 – Present*
Underwater Compliance Robotics Researcher under Professor Daniel Goldman

- Development of an amphibious undulatory limbless robot investigating the role of closed loop integration with mechanical intelligence in open-water navigation and three-dimensional obstacle traversal
- Developed Kalman filter and control loops for pressure sensor and IMU sensor increasing movement speed by 20%
- Implemented asynchronous reading of sensors for cross checking during autonomous navigation
- Implemented finite state machines to ease of teleoperated control
- Awarded Georgia Tech President's Undergraduate Research Award Recipient

Projects

Robot for Autism Behavioral Intervention Humanoid Robot *February 2024*
Engineering Lead under Professor Catherine So

- Engineered the World's Smallest Humanoid Robot certified by Guinness World Records
- Designed and manufactured humanoid leg mechanisms in SolidWorks, optimizing for scale, strength, and manufacturability
- Developed a speech-to-text and text-to-speech model to create conversational AI through API and machine learning integration
- Integrated actuators, embedded compute, sensors, and wiring into a fully functional humanoid platform

World Robot Conference First Robotics Competition *August 2024*
Team Captain

- World Robot Conference First Robotics Competition Winning Alliance Award and Rookie All-Star Award
- Spearheaded the autonomous navigation through development of the Camera Vision using OpenCV in java with 99% Accuracy
- Implemented motion planning in Java using Pure Pursuit to optimize speed and autonomous navigation by 8%
- Analyzed a dataset for robot using differential equations to validate the accuracy of the robot shooter
- Developed math model to graph motor current data in python to optimize speed and accuracy by 15%